

» Air Terminal

[مخرج الهواء]

يوجد Damper على Neck

- ⇒ Square Diffuser
- ⇒ Round Diffuser
- ⇒ Perforated Diffuser
- ⇒ Fresh Air Louver

- ⇒ Slot Diffuser
- ⇒ Swirl Diffuser
- ⇒ Jet Diffuser
- ⇒ Disk Valve

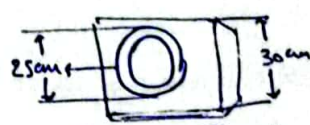
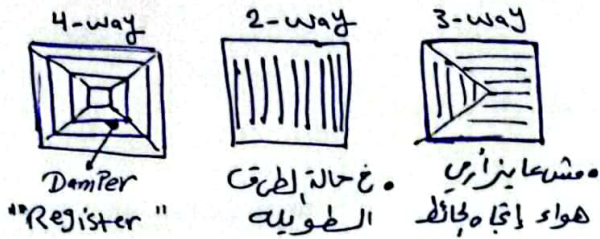
Grille and Register

Door cut and under cut

خصوصاً في الحمامات فنلاحظ ارتفاع باب الحمام حوالي 5:10 cm لدخول الهواء الـ Fresh لأنه ضغط الحمام سالب

Exhausts في الحمامات
نستلها الجاني الأفضل

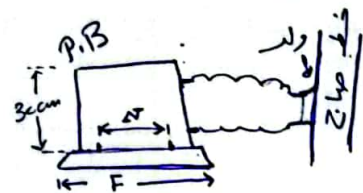
① Square Diffuser



يمكنه انه يكون فتحة Neck في P.B من اساه ولله توصيلة Flexible لازم تكون مفردة لتفليل (losses)

"Plenum Box" علته صاج يتوصل مخرج الهواء بغطر اصاج عنه طريقاً Flexible

التوصيل منه العالي أفضل بسبب التوزيع الأفضل داخل P.B

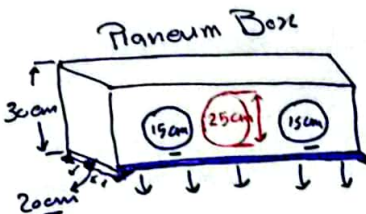
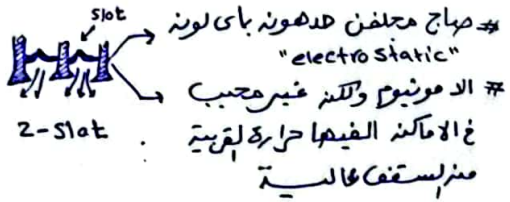


$$F = N + (618)''$$

$$\begin{aligned} \rightarrow F &= 24'' \times 24'' & \rightarrow F &= 24'' \times 24'' & \rightarrow F &= 30'' \times 30'' \\ N &= 18'' \times 18'' & \rightarrow N &= 15'' \times 15'' & \rightarrow N &= 20'' \times 20'' \end{aligned}$$

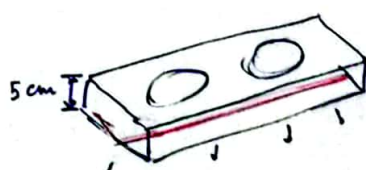
رئيسه Stander له لأنه خفيف غالي يستخدم في الحمامات

② liner Slot Diffuser



لتنظيم ارتفاع Plenum Box همنزود عدد فتحات الرقعة اليك في لزوم الهواء المطلوب "حسب السقف بلناح"

بالأبى حسب وضعيت Slot ننظر مكانه الفتحات



يمكنه عمل الفتحات منه الاعلى

في ال linear slot [1.6] عرض ال (slot) منه [2.5:2] cm



③ Perforated Diffuser

يستخدم في غرف العمليات ويكونه بلاخنة فلتر

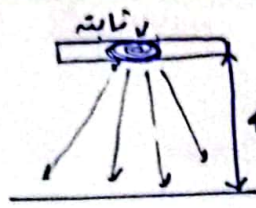
* laminar Air Flow



④ Swirl Diffuser



* ريش ثابت



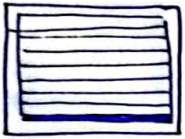
- في حالة أن السقف يبعد عن الأرض حوالي ٤ إلى ٥ متر
يتم استخدام [Swirl Diffuser]
- أم إذا كانت إرتفاع السقف منه ٣ إلى ٤,٥ متر يتم استخدام
الأنواع السابقة

⑤ Jet Diffuser



- في المسافات البعيدة [مخرب الكره] هذه يخرج ترميز لمسافات بعيدة من ١٠ متر
قد تصل إلى ٤٠ متر
- يستخدم المظارات - محطات القطار - مول على ارتفاع عالي و

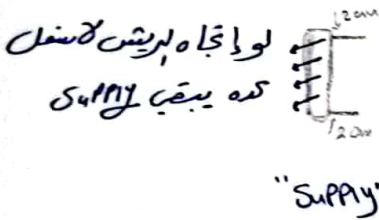
⑥ Fresh Air louver



- يتم تركيبه قبل دخول الهواء Fresh
لمنع دخول الحشرات والظهور

يشبه شيش اشباب

⑦ Grille Diffuser



- أي diffuser يمكن أن يكون [Supply] أو [Return] فلا
في حالة ال grille



ليه هنا يتم وضع الثرموستات عند مخرج [Return] ؟؟؟ « $R \rightarrow T$ »
ولكنه علو طبقة الثرموستات على $T=24^\circ C$ ومقربة مدخل Supply حرارته $12.16^\circ C$ فالبالي
الثرموستات هيفصل اول ما التكييف هيشغل والغرفة هس هتكون مكيفة ف هيفصل هيشغل ويفصل
ام لو وضع جنب مخرج [Return] ف هيفصل عندما تصل حرارة الغرفة لـ $24^\circ C$ مطلوبه.
الممكنه

⑧ Disk Valve

- يوضع في الحمام ليتم من خلاله Exhaust



من المهم هنا كيفية اختيار المخرج ؟
١- بناء على الشكل الجمالي والتنسيق مع المعمارى
٢- عددهم من فلات التحميم وال Sizing

في اختيار مخرج الهواء المناسب لازم

1- تحسب (CFM) Supply ← ، تعد كمية الهواء دي هتخرج منه كام مخرج Exhausted

2- [NC] : تحصل على قيمة معامل لفضوها ر منه خلال جدول Euc صفحة (2) بناء على التقييم
في افضل اختيار بالقيمة [minimum]

3- [velocity] : منه خلال كتاب [Carrier] ← ch2 ← Table 20

Terminal Velocity (F.P.M) ← التقييم

يمكنه تشتغل منه خلال الكود المصري وهو يربط بين مستوى الصوت $Re(N)$ له سرعة الهواء (m/s)

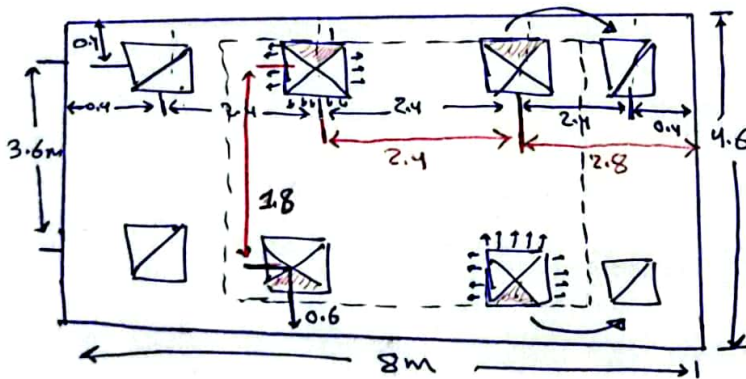
بنحوه فكل سرعة Supply قليلة
سرعة Return عالية

Room From Calculation

4- [Throw]

* $CFM_s = 1600$ CFM

* Assum : 4 Diffuser supply and 4 Diffuser Return



$L = 2.8$
 $L = 2.4$
 $L = 3.8$

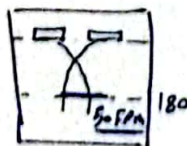
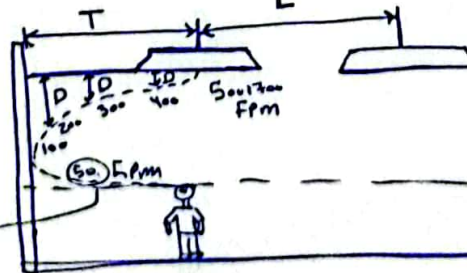
هنا خذ (القيمة)

حتى نضمنه ان باقي ال طول تحت 50 FPM

$$L_{avg} = \frac{L + L + L}{3}$$



يفضل انه
تزيد هذه القيمة عنه
50 FPM



المسافة بين Supply و Return
انه لا تقل عنه 1.8m
حتى لا يحدث short cut
[الصواء ينزل يتم سحبته]



فيه بعض الحالات عاري مثل linear slot
لا غل supply
يتم تحدي اتجاه الرشاة
لا على Return

L → الطول المخرج
 T → مسافة القذف
 D → سقوط الصواء

بناء على نوع المخرج

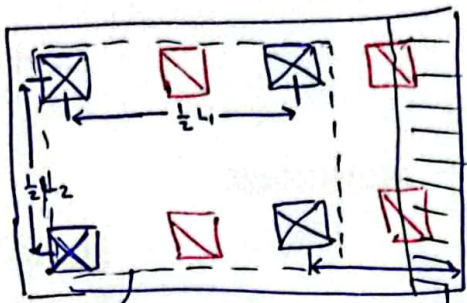
$$T = L * C$$

مفتاح المسافة بين 2-Diffuser
المسافة بين مفتاح المسافة بين 2-Diffuser

الحل هنا في اختيار الطول المناسب

- نستخدم على قيمة P العملية

لضمان انه البعد الجليل السرعة هتقل عن 50 FPM



منطقة تواجد اشخاصه
منطقة تواجد اشخاصه

$$L = 1.8 \text{ m}$$

$$T_{50} = L * C * S.F * 3.28$$



بسبب مقاومة الجاري
التي 24.26°C
للـ Supply

له مسافة القذف التي يكون عندها سرعة الهواء 50 FPM
يتم تحديد بناء على نوع المخرج
ع حالة square diffuser 0.8

$$T_{50} = 1.8 * 0.8 * 1.3 * 3.28 = 6.14 \text{ Ft}$$

معني كدة « انت محتاج مخرج بناء على توزيع المخارج في التصميم

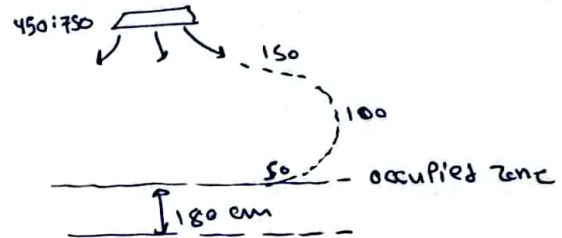
$$\rightarrow \text{No. Diffuser} = 4$$

$$\rightarrow \text{Diffuser} = 400 \text{ CFM}$$

$$\rightarrow \text{NC} = 30$$

$$\rightarrow \text{Velocity} = 450 : 750$$

$$\rightarrow T_{50} = 6.15 \text{ اقل } \downarrow$$



لوحظني بعد مسافة 6.15 او أقل 50 FPM

له دلوقتي همدخل الكتلوجات الخاصة بمخارج الهواء

$$100 * 10 \text{ cm} \rightarrow \text{Bar Grille} \rightarrow 400 \text{ CFM}$$

$$100 * 4 \text{ Slot} \rightarrow \text{linear slot} \rightarrow 200 \text{ CFM}$$





DEFINITIONS:-

- **THROW:** is the horizontal distance between the center of the diffuser and the point at which the terminal velocity of the air stream occurs. (tabulated values of throw in the table represent the minimum and the maximum throw in feet).
- **MINIMUM THROW:** is the distance between the center of the diffuser and the point at which the terminal velocity is 200 (corresponding to a residual velocity of 75 FPM).
- **MAXIMUM THROW:** is the distance between the center of the diffuser and the point at which the terminal velocity is 100 FPM (corresponding to a

residual velocity of 35 FPM). An average value of throw can be taken for general applications such as restaurants, offices, apartments etc. and results in a residual velocity of 55 FPM.

CFM: : the air flow rate in cubic feet per minute.

- **NECK VELOCITY:** is the air stream velocity measured in feet per minute (FPM).
- **TOTAL PRESSURE:** is the sum of static pressure and velocity pressure gives in inches water gauge.
- **NC level(Db):** is the noise criteria based on 10 Db room attenuation and measured at 10 FT away from diffuser face (deduce 3 Db for 20 FT distance and 7 Db for 30 FT distance).
- The sound caused by an air outlets is directly proportional to the velocity of air passing through it (see table 1).

NECK VELOCITY(FPM)	SOUND LEVEL- NC
300-600	Less than 20
700-900	Less than 30
900-1100	Less than 35
1100-1300	Less than 40
1600-1800	Less than 40

Table 1

This table gives the approximate noise level (NC-Value) of a diffuser as a function of the diffuser neck velocity.



RECOMMENDED NOISE CRITERIA FOR ROOMS

TYPE OF AREA	NC LEVEL(Db)	TYPE OF AREA	NC LEVEL(Db)
AUDITORIUMS		SPORT ACTIVITIES INDOOR	
Concert and opera Halls. Studios for sound Reproduction	20-25	Coliseums	30-40
Legitimate theaters. Multi- purpose Halls	25-30	Bowling alleys, gymnasiums	35-45
Movie Theaters. Lecture halls. Planetarium		Swimming pools	40-55
TV audience studios	30-35	TRANSPORTATION	
Lobbies.	35-45	Ticket Sales Offices	30-40
CHURCHES & SCHOOLS		Lounges, waiting rooms	35-50
Sanctuaries	20-30	HOTELS	
Libraries , school and classrooms	30-40	Individual rooms or suits , bell rooms, banquet rooms .	30-40
Laboratories	35-45	Halls & corridors lobbies	35-45
Recreation Halls, corridors and Halls	35-50	Granges, kitchens and laundries	40-50
Kitchens	40-50	MANUFACTURING AREAS	
HOSPITAL AND CLINICS		Foreman's office	40-50
Private rooms	25-35	Assembly lines , light Machinery	55-75
Operating rooms, Wards	30-40	Foundries, heavy Machinery	55-75
Halls, Corridors, laboratories, lobbies and waiting rooms	35-45	OFFICES	
Washrooms and toilets	40-50	Board room	20-30
RESTURANTS & LOUNGES	35-45	Conference Room	25-35
Restaurants	35-50	Tabulation and computation	40-60
Cocktail lounges	35-50	PUBLIC BUILDINGS	
Night clubs	40-50	Public libraries , Museums	30-40
Cafeterias		Post offices, general banking areas, lobbies	35-45
STORES DETAIL	35-45		
Clothing Stores , department stores(upper floors)	40-50		
Department Stores (main floor) supermarkets			

Table 2 Recommended range of noise criteria from ASHREA GUIDE

Table 3 Gives the listed sizes available with the amount of CFM range for each size.

Listed size	CFM range
6" dia.	60-315
8" dia.	105-560
10" dia.	165-870
12" dia.	225-1255
14" dia.	320-1710
16" dia.	420-2240
18" dia.	530-2830
20" dia.	660-3500
24" dia.	940-4950





SUPPLY AIR ROUND CEILING DIFFUSER

PERFORMANCE DATA

Neck Size (inches)	Neck Area (FT ²)	Area factor (FT ²)	Neck Velocity (FPM)	300	400	500	600	700	800	1000	1200	1400	1600
			Velocity Pressure (inches W.G)	0.007	0.011	0.016	0.023	0.031	0.041	0.064	0.091	0.123	0.161
			Air flowrate(CFM)	60	80	100	120	135	155	195	235	275	315
			Total pressure drop	0.025	0.041	0.061	0.082	0.106	0.132	0.201	0.282	0.380	0.480
			Throw (FT)	3-5	3-6	3-6	4-6	4-6	4-6	5-7	5-7	6-8	7-10
			NC	<20	<20	<20	<20	<20	25	30	35	39	45
6"	0.196	0.13	Air flowrate(CFM)	105	140	175	210	245	280	350	420	490	560
			Total pressure drop	0.021	0.032	0.052	0.071	0.100	0.122	0.183	0.271	0.370	0.480
			Throw (FT)	4-6	4-6	5-7	6-8	6-8	6-8	7-10	7-11	8-13	9-14
			NC	<20	<20	<20	20	25	29	33	38	42	46
8"	0.349	0.22	Air flowrate(CFM)	165	220	275	325	380	435	545	655	765	870
			Total pressure drop	0.040	0.061	0.092	0.122	0.181	0.242	0.361	0.500	0.710	0.950
			Throw (FT)	4-6	6-9	7-10	7-10	7-11	8-13	9-14	10-16	12-18	15-22
			NC	<20	<20	<20	21	26	30	34	39	43	47
10"	0.545	0.25	Air flowrate(CFM)	225	315	395	470	550	630	785	940	1100	1255
			Total pressure drop	0.026	0.041	0.072	0.101	0.152	0.201	0.313	0.490	0.670	0.870
			Throw (FT)	6-8	7-10	7-11	7-11	8-13	9-14	10-16	14-21	15-23	19-27
			NC	<20	<20	<20	32	28	32	36	41	45	48
12"	0.785	0.38	Air flowrate(CFM)	320	430	535	640	750	855	1070	1285	1500	1710
			Total pressure drop	0.034	0.051	0.080	0.112	0.156	0.201	0.313	0.460	0.600	0.80
			Throw (FT)	6-8	8-13	9-15	10-16	11-17	12-18	15-23	19-2	19-27	21-31
			NC	<20	<20	<20	24	29	33	37	42	46	49
14"	1.07	0.50	Air flowrate(CFM)	420	560	700	840	980	1120	1400	1680	1960	2240
			Total pressure drop	0.043	0.075	0.106	0.140	0.202	0.262	0.413	0.593	0.800	0.980
			Throw (FT)	7-11	9-14	9-15	11-17	12-18	15-23	16-25	19-27	21-31	25-36
			NC	<20	<20	20	25	30	34	38	43	47	50
16"	1.40	0.59	Air flowrate(CFM)	530	710	885	1060	1240	1415	1770	2125	2480	2830
			Total pressure drop	0.033	0.052	0.082	0.113	0.166	0.220	0.350	0.530	0.710	0.950
			Throw (FT)	8-13	9-14	10-16	12-18	14-12	17-24	19-27	21-31	22-33	26-38
			NC	<20	<20	21	26	31	35	39	44	48	51
18"	1.77	0.75	Air flowrate(CFM)	660	850	1100	1300	1530	1750	2150	2650	3050	3500
			Total pressure drop	0.040	0.065	0.092	0.131	0.182	0.251	0.450	0.640	0.820	0.930
			Throw (FT)	8-14	11-16	14-20	17-21	19-25	22-28	29-34	34-40	41-47	43-50
			NC	<20	<20	21	27	32	36	40	45	49	52
20"	2.18	0.96	Air flowrate(CFM)	940	1250	1570	1850	2200	2500	3130	3850	4320	4950
			Total pressure drop	0.045	0.070	0.105	0.135	0.195	0.265	0.460	0.690	0.790	0.960
			Throw (FT)	10-15	14-20	17-23	20-26	24-30	27-34	35-41	43-50	46-52	51-58
			NC	<20	<20	22	28	33	37	41	46	50	53
24"	3.14	1.23	Air flowrate(CFM)										
			Total pressure drop										
			Throw (FT)										
			NC										

Face [>] 2'4"

قيم ال Throw حسب تحقيق من 6

هذا البراند ← يستخدم في حالة انه قيمة Throw غير مده



RETURN AIR ROUND CEILING DIFFUSER

نلاحظ عدم وجود
Throw

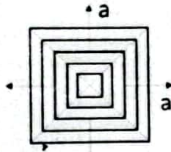
PERFORMANCE DATA

Neck Size (inches)	Neck Area (FT ²)	Neck Velocity (FPM)	300	400	500	600	700	800	1000	1200	1400	1600
		Velocity Pressure (inches W.G)	0.007	0.011	0.016	0.023	0.031	0.041	0.064	0.091	0.123	0.161
6"	0.196	Air flowrate(CFM)	60	80	100	120	135	155	195	235	275	315
		Total pressure drop	0.029	0.048	0.076	0.108	0.146	0.18	0.294	0.421	0.569	0.749
		NC	<20	<20	<20	<20	21	26	32	37	41	46
8"	0.349	Air flowrate(CFM)	105	140	175	210	245	280	350	420	490	560
		Total pressure drop	0.020	0.034	0.053	0.077	0.100	0.123	0.207	0.293	0.394	0.519
		NC	<20	<20	<20	21	26	30	34	39	43	47
10"	0.545	Air flowrate(CFM)	165	220	275	325	380	435	545	655	765	870
		Total pressure drop	0.0101	0.033	0.050	0.071	0.096	0.120	0.196	0.281	0.379	0.477
		NC	<20	<20	<20	22	27	31	36	41	45	48
12"	0.785	Air flowrate(CFM)	225	315	395	470	550	630	785	940	1100	1255
		Total pressure drop	0.018	0.029	0.045	0.064	0.086	0.112	0.176	0.249	0.339	0.443
		NC	<20	<20	<20	23	29	33	37	42	46	49
14"	1.07	Air flowrate(CFM)	320	430	535	640	750	855	1070	1285	1500	1710
		Total pressure drop	0.017	0.028	0.042	0.061	0.083	0.108	0.170	0.242	0.328	0.430
		NC	<20	<20	<20	25	30	34	38	43	47	50
16"	1.40	Air flowrate(CFM)	420	560	700	840	980	1120	1400	1680	1960	2240
		Total pressure drop	0.016	0.025	0.041	0.059	0.079	0.103	0.162	0.231	0.314	0.411
		NC	<20	<20	21	26	31	35	39	44	48	51
18"	1.77	Air flowrate(CFM)	530	710	885	1060	1240	1415	1770	2125	2480	2830
		Total pressure drop	0.015	0.025	0.040	0.057	0.077	0.099	0.156	0.224	0.303	0.397
		NC	<20	<20	22	27	32	36	40	45	49	52
20"	2.18	Air flowrate(CFM)	660	850	1100	1300	1530	1750	2150	2650	3050	3500
		Total pressure drop	0.014	0.024	0.036	0.052	0.071	0.091	0.143	0.205	0.277	0.363
		NC	<20	<20	21	28	33	37	41	46	50	53
24"	3.14	Air flowrate(CFM)	940	1250	1570	1850	2200	2500	3130	3850	4320	4950
		Total pressure drop	0.013	0.022	0.031	0.046	0.064	0.082	0.130	0.185	0.246	0.326
		NC	<20	<20	23	29	34	38	42	47	51	55





PERFORMANCE DATA

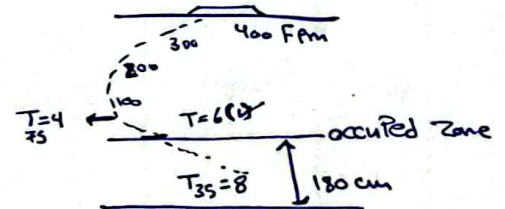


MODEL 4SCD FOUR-WAY THROW SQUARE DIFFUSER

NECKSIZE AK	Neck VELOCITY FPM	200	300	400	500	600	700	800	900
	TOTAL PRESSURE	.019	.039	.067	.10	.14	.19	.24	.30
6"x6"	TOTAL CFM	50	75	100	125	150	175	200	225
AK= 0.13	CFM each side - a	13	19	25	32	38	44	50	57
	Throw each side - a	2-3	3-5	4-7	5-9	6-11	7-13	8-15	9-17
9"x9"	TOTAL CFM	112	168	224	280	336	392	448	504
AK= 0.28	CFM each side - a	28	42	56	70	84	98	112	126
	Throw each side - a	2-3	3-5	4-8	5-10	6-12	7-14	9-17	9-18
12"x12"	TOTAL CFM	200	300	400	500	600	700	800	900
AK= 0.5	CFM each side - a	50	75	100	125	150	175	200	225
	Throw each side - a	2-4	4-7	5-9	6-11	7-14	8-16	10-19	11-21
15"x15"	TOTAL CFM	312	468	624	780	936	1092	1248	1404
AK= 0.79	CFM each side - a	78	117	156	159	235	273	312	351
	Throw each side - a	3-5	4-8	5-10	7-13	8-16	10-19	11	12-24
18"x18"	TOTAL CFM	450	675	900	1125	1350	1575	1800	2025
AK= 0.13	CFM each side - a	113	169	225	281	338	394	450	507
	Throw each side - a	3-5	4-8	5-10	7-13	9-17	10-20	12-24	13-26
21"x21"	TOTAL CFM	612	918	1224	1530	1836	2142	2448	275
AK= 1.54	CFM each side - a	153	229	306	382	459	535	612	689
	Throw each side - a	3-5	4-8	6-11	7-13	9-18	11-21	13-25	14-28
24"x24"	TOTAL CFM	800	1200	1600	2000	2400	2800	3200	3600
AK= 2	CFM each side - a	200	300	400	500	600	700	800	900
	Throw each side - a	3-6	5-9	6-12	7-14	10-19	11-22	13-26	15-30

→ F > [6:10]"

- ☐ Diffuser selections which will result in noise criteria not exceeding
Approximately 20 DB based on room attenuation OF 8 DB
- ☒ Resulting criteria not exceeding 30 DB
- ☐ Resulting criteria not exceeding approximately 40 DB.
- ☐ Resulting criteria exceeding 40 DB.



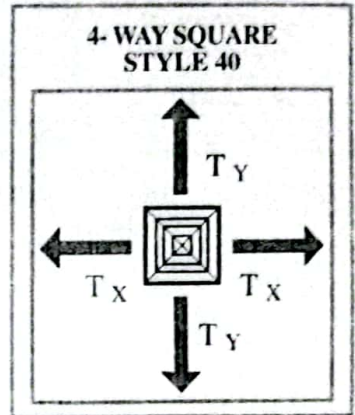
Square Diffuser



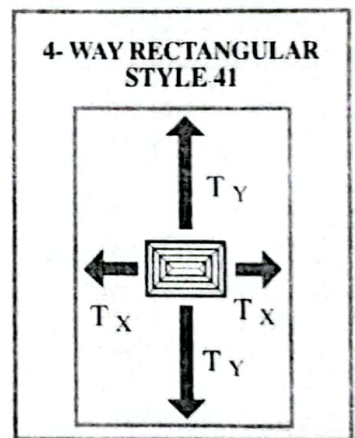
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4 WAY ENGINEERING PERFORMANCE DATA

Listed Size Outlet Area		V ₀ Outlet Velocity FPM											
		500	600	700	800	900	1000	1200	1400	1600	1800	2000	
		P _t Total Pressure Inches H ₂ O											
		.02	.02	.03	.04	.05	.06	.09	.12	.16	.20	.25	
6 x 6 A ₀ .10	CFM	50	60	70	80	90	100	120	140	160	180	200	
	T X Y	2-3 2-3	2-3 2-3	2-4 2-4	2-4 2-4	3-5 3-5	3-5 3-5	4-6 4-6	4-8 4-8	5-8 5-8	5-9 5-9	6-11 6-11	
9 x 9 A ₀ .22	CFM	110	135	155	180	205	225	270	315	360	410	450	
	T X Y	2-4 2-4	2-4 2-4	3-5 3-5	3-5 3-5	4-6 4-6	5-8 5-8	5-9 5-9	6-11 6-11	6-12 6-12	7-13 7-13	8-14 8-14	
12 x 12 A ₀ .40	CFM	200	240	280	320	360	400	480	560	640	725	800	
	T X Y	3-5 3-5	4-6 4-6	4-8 4-8	5-8 5-8	5-9 5-9	6-11 6-11	6-12 6-12	7-13 7-13	8-15 8-15	9-17 9-17	10-19 10-19	
15 x 15 A ₀ .62	CFM	310	375	440	500	565	625	750	875	1000	1125	1250	
	T X Y	4-6 4-6	4-8 4-8	5-9 5-9	6-11 6-11	6-11 6-11	6-12 6-12	8-15 8-15	10-18 10-18	10-19 10-19	12-21 12-21	13-23 13-23	
18 x 18 A ₀ .90	CFM	450	540	630	720	810	900	1080	1260	1440	1620	1800	
	T X Y	4-8 4-8	5-9 5-9	5-11 5-11	6-12 6-12	7-13 7-13	8-15 8-15	10-17 10-17	11-20 11-20	13-23 13-23	15-27 15-27	16-30 16-30	
21 x 21 A ₀ 1.23	CFM	615	740	860	985	1110	1230	1475	1725	1970	2220	2460	
	T X Y	5-9 5-9	6-11 6-11	7-13 7-13	8-14 8-14	9-15 9-15	9-17 9-17	11-21 11-21	13-25 13-25	15-29 15-29	17-31 17-31	19-35 19-35	
24 x 24 A ₀ 1.6	CFM	800	960	1120	1275	1440	1600	1925	2240	2570	2890	3200	
	T X Y	5-11 5-11	7-13 7-13	7-14 7-14	8-15 8-15	9-17 9-17	10-19 10-19	12-23 12-23	14-29 14-29	16-31 16-31	18-35 18-35	20-39 20-39	
27 x 27 A ₀ 2.02	CFM	1010	1215	1420	1615	1820	2020	2430	2840	3240	3650	4060	
	T X Y	6-12 6-12	7-13 7-13	8-15 8-15	10-18 10-18	10-19 10-19	12-22 12-22	14-27 14-27	16-32 16-32	18-35 18-35	20-38 20-38	23-42 23-42	
30 x 30 A ₀ 2.50	CFM	1250	1500	1750	2000	2250	2500	3000	3500	4000	4500	5000	
	T X Y	7-13 7-13	8-14 8-14	9-16 9-16	11-19 11-19	11-20 11-20	13-23 13-23	15-28 15-28	17-32 17-32	19-36 19-36	21-39 21-39	24-43 24-43	
33 x 33 A ₀ 2.75	CFM	1370	1650	1925	2200	2470	2750	3300	3850	4400	4950	5500	
	T X Y	7-13 7-13	9-16 9-16	10-18 10-18	12-21 12-21	14-24 14-24	16-27 16-27	18-33 18-33	19-37 19-37	23-41 23-41	27-46 27-46	31-50 31-50	
36 x 36 A ₀ 3.60	CFM	1800	2160	2520	2880	3240	3600	4320	5040	5760	6480	7200	
	T X Y	9-15 9-15	10-17 10-17	11-19 11-19	13-22 13-22	15-25 15-25	17-28 17-28	19-34 19-34	20-38 20-38	24-42 24-42	28-47 28-47	32-51 32-51	



	≥ NC45
	NC40
	NC35
	NC30
	≤ NC30



	≥ NC45
	NC40
	NC35
	NC30
	≤ NC30

Note: The minimum T Dimension is based on a V₀ of 200 FPM with a V₀ of 65 FPM. The maximum T Dimension is based on a V₀ of 100 FPM with a V₀ of 15 FPM.

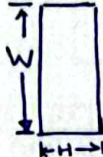
4- قیمة Throw





ENGINEERING PERFORMANCE DATA

TABLE - 1

(h) LISTED HEIGHT (in) 					Outlet Velocity 450-600 FPM				Outlet Velocity 550-700 FPM				Outlet Velocity 600-800 FPM							
					CFM	Deflection				CFM	Deflection				CFM	Deflection				
						0°	22°	42°	55°		0°	22°	42°	55°		0°	22°	42°	55°	
						V _k					V _k					V _k				
						450	500	550	600		550	600	650	700		600	700	750	800	
Pressure P _t Dcol					P _t					P _t										
.01 .01 .02 .02					.02 .02 .03 .03					.02 .03 .04 .04										
Throw (T) V _t 75					Throw (T) V _t 75					Throw (T) V _t 75										
4	5	6	8, 10 12	14 16 18	20 24	40	6	5	4	3	45	8	7	6	4	60	10	9	7	5
6	5					50	7	6	5	4	60	9	11	6	5	70	11	10	8	6
10	8	6				75	9	7	6	5	90	10	9	7	6	100	12	11	9	7
12	10	8				100	10	8	6	5	125	11	9	8	7	150	13	12	9	8
14	12	10	8			125	11	9	7	6	150	12	11	9	7	175	14	13	10	8
18	14	12				150	11	10	8	7	175	13	11	10	8	200	15	14	11	9
20	16	14	10			175	12	11	9	7	200	15	13	11	9	250	17	15	12	9
24	20	16	12 10			225	13	12	10	8	275	16	14	12	10	300	19	16	13	10
30	24	20	14 12			300	14	13	11	9	350	18	15	13	10	425	20	17	15	11
38	30	24	18 14 12			350	17	14	12	9	425	20	17	13	11	500	22	20	17	12
40	32	26	20 16 14			400	18	16	13	10	475	21	18	15	12	550	24	21	18	13
44	36	30	24 18 16	14		450	19	17	13	11	550	22	19	17	13	625	26	22	19	14
44	36		26 22 18	16		500	20	17	14	12	600	24	20	18	13	700	27	24	20	15
48	40		30 24 20	18 16		600	21	18	15	12	700	25	22	18	14	850	29	25	22	16
48			38 30 24	22 18		700	23	20	17	13	850	28	24	20	15	975	33	28	24	18
			40 32 28	24 20 18		800	25	21	18	14	950	31	26	22	17	1125	35	30	26	20
			42 36 30	26 24 22	20	900	27	22	19	15	1100	32	27	23	18	1250	36	31	27	21
			46 42 36	30 26 24	22	1000	29	25	21	16	1200	34	29	25	19	1400	40	34	29	22
			44	38 34 30	28 24	1250	32	27	23	18	1500	36	30	27	23	1750	43	37	31	24
				48 42 36	34 28	1500	34	29	25	19	1800	39	32	29	25	2100	45	40	34	26
				44 40	36 30	1750	37	32	27	21	2100	42	35	32	27	2450	48	44	37	28
				48	42 36	2000	39	34	29	22	2400	45	38	34	29	2800	52	46	40	30
					48 40	2250	41	36	30	24	2700	48	40	36	31	3150	54	48	42	32
					42	2500	43	38	32	25	3000	51	42	38	33	3500	57	51	44	33
					48	3000	46	40	34	27	3600	54	45	40	35	4200	60	53	47	35

SYMBOLS

V_t Terminal Velocity in FPM

V_R Room Velocity in FPM

V_k Outlet Velocity in FPM

A_k Outlet Area in Sq. Ft

P_t Total Pressure in. H₂ O

P_s Static Pressure in. H₂ O

NC re 18db room attenuation

T Throw in feet

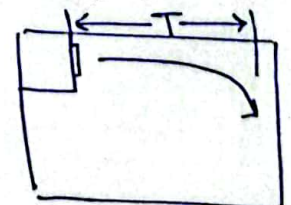
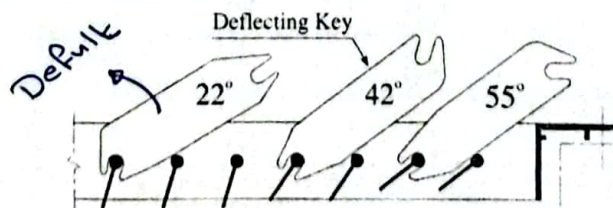
	≤ NC30
	NC35
	NC40
	≥ NC45

The indicated air paths may be easily obtained by using the appropriate notch on the bar deflecting key.
The bars should be adjusted as shown for the required pattern

$$Q = A * V$$

$$A = \frac{Q}{V}$$

$$A = \frac{Q}{V} \rightarrow \text{in}^2 \rightarrow \frac{\pi}{4} D^2 = A$$





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400 CFM حدود 1م
حدود بطول بطول 1م

Linear
Grille

ENGINEERING PERFORMANCE DATA

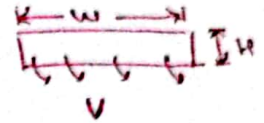
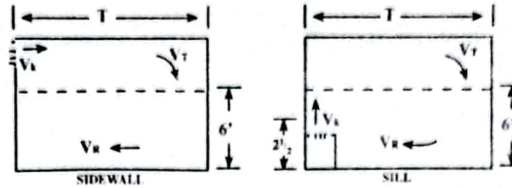


Table 1- Supply Air*

CFM Per Foot	Listed Height in Inches	Min. Ps In. H ₂ O	Face Velocity (V _f) FPM	Throw (T) in Feet		Minimum Ceiling Height in Feet		NC
		Bar Style	Bar Style	Sidewall	Sill	@ - 18F ΔT	@ - 25F ΔT	
		00 and 15	00 and 15	Min. - Max.	Min. - Max.			
20	1 1/2	.01	500	6-9	1-2	8	9	<20
30	1 1/2	.03	750	7-10	2-3	9	10	25
	2	.01	475	6-9	1-2			20
40	1 1/2	.05	1000	9-13	3-5	9	11	30
	2	.02	635	8-11	2-4			25
	2 1/2	.01	460	7-10	2-3			20
50	1 1/2	.09	1250	11-16	4-9	9 1/2	11	30
	2	.03	790	10-14	3-7			25
	2 1/2	.02	575	9-13	2-6			20
	3	<.01	440	8-12	2-5			<20
60	2	.05	950	12-18	5-11	9 1/2	12	30
	2 1/2	.02	690	11-16	4-9			25
	3	.01	530	10-14	3-7			20
	4	<.01	370	8-12	2-5			<20
70	2	.06	1110	14-20	7-13	10	12	30
	2 1/2	.03	810	13-19	6-12			25
	3	.02	660	11-16	4-9			20
	4	<.01	435	10-14	3-7			<20
80	2	.08	1275	16-23	9-16	10 1/2	12 1/2	30
	2 1/2	.04	920	15-21	8-14			25
	3	.03	700	13-18	6-11			20
	4	.01	495	11-16	4-9			<20
90	2 1/2	.05	1030	17-24	10-17	11	13	30
	3	.04	785	15-21	8-14			25
	4	.01	550	13-18	6-11			20
	5	<.01	450	11-16	4-9			<20
100	2 1/2	.06	1150	19-27	12-20	11	13	30
	3	.04	875	16-23	9-16			25
	4	.02	620	14-20	7-13			20
	5	.01	500	12-18	5-11			<20
120	3	.06	1050	19-28	11-20	11 1/2	13	30
	4	.03	745	17-24	9-16			25
	5	.02	600	15-22	7-14			20
	6	<.01	480	13-19	5-11			<20
140	3	.08	1220	22-32	14-24	11 1/2	14	35
	4	.04	870	19-28	11-20			25
	5	.02	700	17-25	9-17			20
	6	.01	560	15-22	7-14			<20
160	4	.05	990	22-32	13-23	12	15	35
	5	.03	800	19-29	10-20			25
	6	.02	640	18-26	9-17			20
	8	.01	460	15-22	6-13			<20
180	4	.07	1110	25-36	16-27	12	15	35
	5	.04	900	22-33	13-24			25
	6	.03	725	20-30	11-21			20
	8	.02	520	17-25	8-16			<20
200	4	.08	1240	28-41		12	15	40
	5	.05	1000	24-36				35
	6	.04	800	23-33				30
	8	.02	575	20-28				25
250	5	.08	1250	30-46		13	15	40
	6	.05	1000	27-39				35
	8	.03	720	25-35				30
	10	.01	550	21-32				25
300	6	.07	1200	33-48		13	15	40
	8	.04	865	29-42				35
	10	.02	665	25-39				30
	12	.01	545	23-33				25
350	8	.05	1020	34-48		13	15	40
	10	.03	780	29-45				35
	12	.02	640	26-38				30
400	8	.08	1170	40-55		14	16	45
	10	.04	890	33-50				40
	12	.03	730	30-44				35

Symbols : V_t Terminal Velocity in FPM
V_r Room Velocity in FPM
V_f Face Velocity in FPM

A_o Outlet Area in Sq. Ft.
A_n Neck Area in Sq. Ft.
P_s Static Pressure inches H₂O.

N_r re 18db Room Attenuation
T Throw in Feet, see note f.
ΔT Temperature Differential



Linear Slot



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ENGINEERING PERFORMANCE DATA

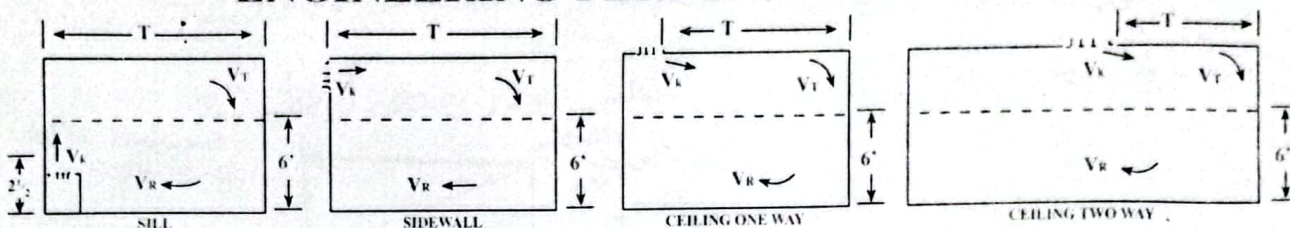


TABLE 1 SUPPLY AIR :

CFM Per Foot In Direction of T	No. Of Slots	Min. Ps In. H ₂ O	Outlet Velocity (V _n) FPM	Throw (T) in Feet			Minimum Ceiling Height in feet		NC
				Ceiling	Sidewall	Sill	@ - 18F-ΔT	@ - 25F-ΔT	
				Min. - Max.	Min. - Max.	Min. - Max.			
10	1	.01	335	4-6	2-4	1-2	7 1/2	9	<20
20	1	.04	670	8-11	6-9	2-3	8	9	20
	2	<.01	400	6-9	4-7	1-2			<20
30	1	.09	1000	10-14	8-12	3-4	9	10	25
	2	.02	600	8-11	6-9	2-3			20
	3	<.01	430	7-9	5-7	1-2			<20
40	1	.16	1340	13-17	11-15	4-6			30
	2	.04	800	10-14	8-12	3-4	9	11	25
	3	.02	575	9-12	7-10	2-3			20
	4	.01	445	8-11	6-9	2-3			<20
50	2	.06	1000	11-15	9-13	4-6			25
	3	.03	715	10-14	8-12	3-4	9 1/2	11	20
	4	.02	555	9-13	7-11	2-4			<20
	5	<.01	415	7-12	6-10	2-3			<20
60	2	.09	1200	13-17	11-15	5-8			30
	3	.04	860	12-16	10-14	4-7	9 1/2	12	25
	4	.02	665	11-15	9-13	3-6			20
	5	.01	500	9-13	7-11	3-4			<20
70	2	.13	1400	15-20	13-18	6-11			30
	3	.06	1000	13-18	11-16	5-9	10	12	25
	4	.03	775	12-16	10-14	4-7			20
	5	.02	585	10-15	8-13	3-5			<20
	6	.01	500	9-14	7-12	2-5			<20
80	3	.07	1140	14-20	12-18	6-11			30
	4	.04	885	13-19	11-17	5-10	10 1/2	12 1/2	25
	5	.03	665	13-17	11-15	4-8			20
	6	.02	575	12-16	10-14	3-7			<20
	7	<.01	500	11-15	9-13	3-6			<20
90	3	.09	1290	17-24	15-21	8-14			30
	4	.05	1000	16-22	14-20	7-13	11	13	25
	5	.03	750	15-20	13-18	6-11			20
	6	.02	645	14-18	12-16	5-9			20
	7	.01	560	13-17	11-15	4-8			<20
100	3	.13	1430	19-26	17-23	10-16			35
	4	.06	1110	18-25	16-22	9-15	11	13	30
	5	.04	830	16-23	14-20	7-13			25
	6	.03	715	14-20	12-18	6-11			20
	7	.02	630	13-19	11-17	5-10			<20
120	4	.09	1330	19-27	17-24	10-16			30
	5	.06	1000	18-26	16-23	8-15	11 1/2	13	25
	6	.04	860	17-25	15-22	7-14			20
	7	.03	750	16-23	14-20	6-12			20
	8	.02	630	15-20	13-18	5-10			<20
140	5	.08	1170	20-30	18-27	10-19			30
	6	.06	1000	19-28	17-25	9-17	11 1/2	14	25
	7	.04	875	18-26	16-23	8-15			25
	8	.03	740	16-24	14-21	6-13			20
160	6	.07	1150	21-32	19-29	10-20			25
	7	.05	1000	20-30	18-27	9-18	12	15	25
	8	.04	840	18-27	16-24	8-16			20
180	6	.09	1290	24-35	21-31	12-22			30
	7	.07	1130	23-34	20-30	11-21	12	15	30
	8	.06	950	20-31	18-28	9-19			25
200	6	.11	1440	26-40	23-36	-			30
	7	.08	1250	25-38	22-34	-	12	15	30
	8	.06	1110	24-36	21-32	-			25
250	8	.10	1315	26-46	23-41	-	13	15	35

SYMBOLS V_t Terminal Velocity in FPM
V_r Room Velocity in FPM
V_n Outlet Velocity in FPM

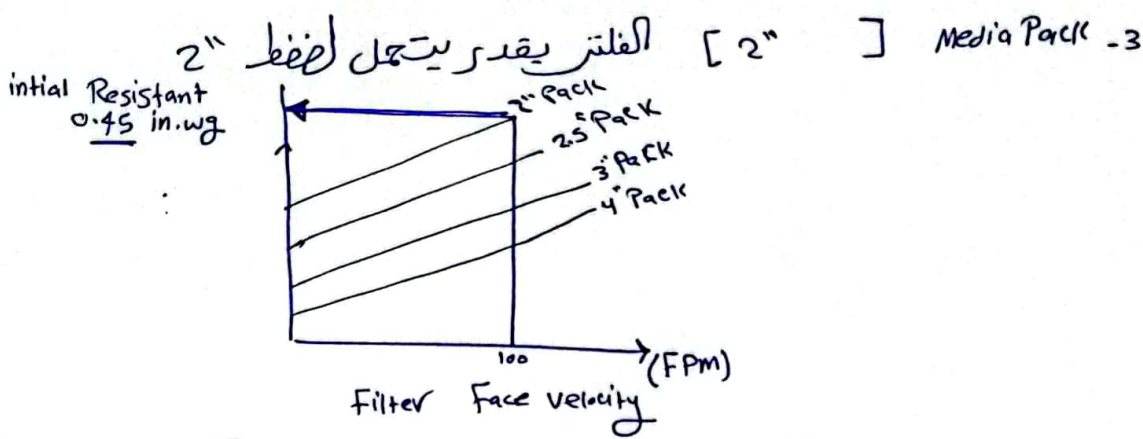
A_n Outlet Area in Sq. Ft.
A_n Neck Area in Sq. Ft.
P_s Static Pressure inches H₂O

N_r re 18db Room Attenuation
T Throw in Feet
ΔT Temperature Differential



LAF Selection

- كيفية عمل selection للفلتر داخل مخرج الهواء « لانه كمية الهواء المتدخل في غرفة لازم نعرفها من الفلتر »
- « هنتعرف 1- Actual size [$24 \times 24 \times 2 \frac{3}{4}$]
- « 2- Air Flow Rate [350 CFM] ولكنه هتصمم على CFM اقل منه ال supply
- ولكنه مثلا 165 cfm



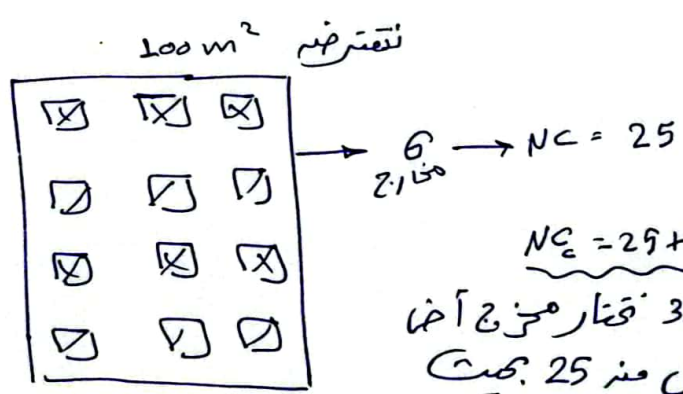
كل ما الارتفاع زاد كل ما قيمة [media Pack] ~~تزيد~~ ~~تقل~~ ~~تزيد~~ كلما Initial Resistance تقل --

« بنعمل selection بناءً على كمية CFM المخرج منه »

في حالة مساحة الغرفة كبير و يوجد عدد كبير من المخرج فيمنه اعمار مساحة تنامح منه

فقط من اصل مساحة الاجمالية لتطبيق قيمة المخرج علي

→ No. of outlets	1	2	3	4	8	10	20	40
→ dB Boost	0	3	5	6	9	10	13	16



$NC_e = 25 + 7 = 32$

غير مقبول $NC = 32$ فقنا مخرج آخر

لكن NC له اقل منه 25 بحيث

لوجعنا 7 يبقى NC_e قليل



فوقه لوساحة الكبر منه

سده هنتطبق لوافل في تطبيق

